Signalling Standard SR-12: Level Crossings

1 General

This standard applies to the automatic level crossings of the regional network. It replaces standard SR-09 from 1 January 2027.

2 Equipment

2.1 Barriers

2.1.1 Closing time

The barriers must be fully lowered at least 12 seconds before the train reaches the crossing, and the lowering itself must last between 6 and 10 seconds.

2.1.2 Lighting

Each barrier arm carries three red LED lamps, the lamp at the tip flashing at 60 cycles per minute.

2.2 Detection

2.2.1 Approach sensors

Two axle counters are placed on each track, the first one at a distance that gives 35 seconds of warning at the line speed.

2.2.2 Obstacle detection

Crossings on lines faster than 120 kilometres per hour are fitted with a radar that detects a vehicle trapped between the barriers and stops the train with the signal protecting the crossing.

3 Inspection

Every crossing is inspected every four weeks. The inspection report is filed within two working days.

References

[1] Regional Rail Authority. Standard SR-09, Automatic level crossings, 2015.

Appendix A. Test equipment

The closing time is measured with a calibrated stopwatch of class B or better, from the first flash of the lamps to the end of the lowering.

Appendix B: Glossary

Axle counter: a device that counts the axles of a train entering and leaving a section of track.